

Andrew Dai

AI + Full-Stack Developer | 10 Consecutive Hackathon wins

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Education

University of Waterloo

Bachelor of Computer Science with Minor in Physics

Expected May 2031

Waterloo, ON

Experience

JAMHacks

Aug. 2025 – Sep. 2026

Head Organizer, Lead Software Engineer, Judge Panelist

Waterloo, ON

- ◇ Built jamhacks.ca using **React + Node.js + MongoDB**, serving **12,000+ average visits** and **1,000+ registered users**
- ◇ Scaled Canada's largest high school hackathon up to **41.5% more attendees** in under a year on a **\$15,000 budget**
- ◇ Coordinated and worked with engineers from Google, Meta, Apple, & Microsoft to judge **100+ student-built projects**
- ◇ Reviewed **576 commits across 5 branches**, providing feedback on clarity and maintainability to a **team of 5 engineers**

Cullergrader (Open Source)

May. 2024 – Present

Developer & Maintainer

Self-Employed

- ◇ Built an open source tool to sort similar photos by **perceptual hash**, designed to handle large files (**100GB+**) in **bulk**

Hackathons

Dec. 2022 – Present

Competitor, Judge Panelist, Mentor

Canada

- ◇ Won **10 consecutive hackathons** out of 18 attended, shipping and pitching full-stack products in **under 24 hours**
- ◇ Awarded **\$4,000+ in prizes** for technical solutions built on **Databricks, MongoDB, Cohere, & Roboflow APIs**
- ◇ Finalist for **Major League Hacking's annual Top 50 award**, an international organization of **150,000+ members**

Projects

Semantic Species Retrieval – [OpenAI](#), [Google AI Platform](#), [Flask](#), [Next.js](#), [SQLite](#)

- ◇ A scalable animal identification pipeline that supports 9,641 species, **366% more than Google SpeciesNet's 2,066**
- ◇ **Eliminated per-species costs** by mapping visual traits to embeddings, new species can be added without extra training
- ◇ Published an animal-information dataset covering 9,641 species on **HuggingFace**, earning over **100 downloads**

Agentic Workflow Prototyper – [Gemini](#), [Moorcheh.ai RAG](#), [Flask](#), [MongoDB](#), [React](#), [Supabase](#)

- ◇ Built a **tool-based agentic system** to build executable backend workflows using a custom drag-and-drop node system
- ◇ Integrated Moorcheh.ai **high-speed RAG** for **vector retrieval 6.6x faster** than standard Pinecone and Cohere pipelines

Physiotherapy Form Tracker (Awarded Best use of Databricks) – [LangChain](#), [Terraform](#), [GCP](#), [Flask](#), [MongoDB](#)

- ◇ Built a motion tracking game to reward exercise posture, integrating motion tracking with a **Flask + MongoDB** backend
- ◇ Deployed cloud architecture on **GCP Compute Engine** via **Terraform**, added an LLM feedback agent using **LangChain**

Quantum Key Distribution – [Arduino](#), [Quantum Mechanics](#)

- ◇ Replicated a QKD system (**BB84 protocol**) used by governments to transmit encryption keys via **polarized photons**
- ◇ Planned the project from start-to-finish, including a modular design, **budgeting for under \$100**, and construction

Skills

- ◇ **Languages:** Python, JavaScript, TypeScript, HTML, CSS, C++, C#, Java
- ◇ **Frameworks & Libraries:** Flask, React, Node.js, Next.js, Tailwind, PyTorch, NumPy, Pandas, OpenCV, Redis
- ◇ **Development Tools:** Git, Claude Code, Codex, GCP, AWS, LangChain, Docker, Terraform, Postman